FPT UNIVERSITY

CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Can Tho, November 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
12/09/2026	A	Pham Kha Vy	Initial draft of the Software Test Documentation: testing scope, test levels, constraints, and test strategy.
03/10/2026	A	Nguyen Tien Thuan	Added unit test and system test plans for Sprint 1 (User Authentication, Role Management, Profiles, and Job Marketplace).
17/10/2026	M	Tran Phat Tai	Updated integration and system test specifications for Sprint 2 (Proposals, Contracts, Milestones, Permissions, Sessions, and Forum Categories).
24/10/2026	M	Nguyen Lu Nhat Huy	Added test specifications for Sprint 3 (Real-time Messaging, Notifications, Community Forum, Moderation, ID Verification, and User Management).
31/10/2026	M	Vo Nhut Tin	Added test specifications for Sprint 4 (Stripe Escrow Payments, Webhooks, Disputes, Advertisement Banners, and Skill Management).
02/11/2026	M	Nguyen Tien Thuan	Added Sprint 5 test specifications (Job Category Management, AI CV Analysis, Expert Directory) and completed unit test sheets `UF-01` to `UF-84`.
04/11/2026	M	Pham Kha Vy	Consolidated final test execution summary (507 unit test cases and 411 system test cases), defect metrics, test analysis, and conclusion.

**A - Added M - Modified D - Deleted*

II. Testing Documentation

1. Scope of Testing

1.1 Testing Scope

1.1.1 In-Scope Testing

Testing for the **Frevia – Verified Freelance Marketplace Platform** (Project Code: `SEP490_04`) covers all functional and non-functional deliverables across the five project sprints:

- **Identity, Authentication, and Access Control (FE-01 to FE-09):**
  - Email and password registration with one-time-code (OTP) email verification, Google OAuth 2.0 sign-in, and linking a Google identity to an existing account.
  - JWT access and refresh token authentication, persisted session management (listing active sessions, viewing device/IP details, and revoking individual sessions), password reset via OTP, and password change.
  - Multi-role membership (`CLIENT`, `FREELANCER`, `ADMIN`, and custom administrative roles), joining additional roles, active session role switching, and dynamic data-driven RBAC permission validation (`HTTP method + path + module`).
- **User Profiles, Portfolios, and AI CV Analysis (FE-10 to FE-20):**
  - Account profile management with avatar/cover uploads, freelancer profiles (professional title, bio, education, certifications, languages, work experience, skills with proficiency levels 1–10, and portfolio items), client company profiles, and expert directory profiles.
  - Profile completion and strength scoring, administrative approval workflow for profile revisions, PDF curriculum vitae upload (up to 10 MB), and AI-assisted CV skill extraction.
- **Electronic Identity Verification / eKYC (FE-21 to FE-23):**
  - Submission of front/back identity card images, selfie capture, and active/passive liveness frames; AI verification pipeline (OCR, QR code reading, face matching, tamper detection); and the administrative review queue.
- **Job Marketplace, Proposals, Contracts, and Milestones (FE-24 to FE-47):**
  - Fixed-price job creation, editing, soft deletion, status lifecycle, stable slug generation, multi-criteria public job search, job bookmarks, and saved searches.
  - Proposal drafting, submission, editing, withdrawal, and expiry; client proposal evaluation and hiring decision; dual-signature contract execution; USD 10.00 platform fee tracking; and milestone creation, funding state transitions, work submission, change requests, and approval.
- **Stripe Escrow Payments, Disputes, and Reviews (FE-48 to FE-65):**
  - Stripe Connect Express onboarding, saved payment methods, platform fee and milestone funding via `PaymentIntent` and `Checkout Session`, milestone transfers on release, refunds, raw-body webhook signature verification, and ledger reconciliation.
  - Milestone dispute filing with evidence, respondent defence within deadlines, decision review escalation, administrative arbitration with split payouts, and post-contract reviews/ratings.
- **Real-Time Communication, Community Forum, and Administration (FE-66 to FE-83):**

- Authenticated Socket.IO WebSocket messaging (1-on-1 conversations, file attachments, read receipts, typing indicators, pin/hide/delete) and persistent notifications.
- Community forum browsing, rich-text post creation (Tiptap), comments, likes, reporting, AI moderation scoring, and administrative moderation/trash management.
- Administrative management of users, roles, permissions, sessions, job categories, master skill catalogue (soft-delete and restore), and advertisement banners across five placement zones, together with bilingual localization (`en`/`vi`) and responsive light/dark themes.

1.1.2 Out-of-Scope Testing

In accordance with the project scope limitations defined in Report 1, the following areas are excluded from testing:

- **Third-Party Core Infrastructure Internals:** Internal uptime, cryptographic engines, and settlement clearing networks of external service providers (Stripe live banking rails, Google OAuth identity servers, Cloudinary CDN infrastructure, and external SMTP servers).
- **Hourly Contracts and Multi-Currency Settlement (LI-01 to LI-04):** Hourly billing, automated time-tracking diaries, non-USD currency conversion, domestic e-wallets, direct bank wire transfers, and cryptocurrency settlement.
- **Machine Learning Model Training and Foreign ID Documents (LI-05 to LI-07):** Training or fine-tuning underlying neural network weights for OCR, liveness, or CV parsing, as well as identity verification for non-Vietnamese passports or driver's licences.
- **Native Mobile Clients and Multi-Replica Clustering (LI-09, LI-13):** Native iOS/Android mobile applications, multi-node horizontal load balancing, sticky session clustering, and distributed Redis WebSocket adapters.
- **Unexposed Database Entities (LI-11):** Schema tables reserved for future releases that are not exposed through the current API or UI (`totp_secrets`, `invoices`, `certificates`, `contract_terms_templates`, `job_alerts`, `banned_keywords`, and `system_settings`).

1.2 Testing Levels

Unit Test → Integration Test → System Test

1.2.1 Unit Testing

- **In-charge:** All Developers (`Nguyen Tien Thuan`, `Vo Nhut Tin`, `Tran Phat Tai`, `Nguyen Lu Nhat Huy`), reviewed by `Pham Kha Vy`.
- **Input / Time:** Software Design Document (Report 4), shared Zod validation schemas (`@shared/types`), and NestJS service/repository/guard source code; executed continuously throughout Sprints 1–5 prior to creating pull requests.
- **Focus:**
  - Isolated business logic in NestJS services (`ForumService`, `ConversationService`, `ModerationService`, `IdVerificationService`, `AuthService`, `RolesService`, `PermissionsService`, `SessionsService`, `ForumAdminService`, `UsersService`, `ExpertProfileService`, `BannersAdminService`,

SkillsAdminService, JobCategoriesAdminService, PaymentService, DisputeService, and ProfileService).
  - Data access queries and transaction callbacks in repository classes using mocked Prisma clients.
  - Token verification, active role resolution, and route-level RBAC permission checking in AccessTokenGuard.
  - Normal execution paths (N), abnormal exception branches (A – 400, 401, 403, 404, 409, 422), and boundary value conditions (B).
- **Tools:** Jest 30.0.0, @nestjs/testing 11.0.1, ts-jest 29.2.5, TypeScript 5.9.2, ESLint 9.18.0.
- **Acceptance Criteria:**
  - 100% of the 507 documented unit test cases across 84 service functions (UF-01 to UF-84) pass with zero failures.
  - At least 80% statement and branch coverage on critical domain services and guards.
  - pnpm lint, pnpm check-types, and pnpm --filter api test pass without errors.

1.2.2 Integration Testing

- **In-charge:** Tran Phat Tai, Vo Nhut Tin, and Nguyen Lu Nhat Huy, supported by Nguyen Tien Thuan and reviewed by Pham Kha Vy.
- **Input / Time:** Merged feature branches on the integration environment; executed upon completion of each functional module across Sprints 1–5.
- **Infrastructure:** Docker Compose stack running NestJS 11 API (localhost:3002), Next.js 16 Web Client (localhost:3003), PostgreSQL 16 with Prisma 7 migrations, Stripe Test Mode sandbox, and local/mocked AI HTTP services.
- **Focus:**
  - Controller-to-Service-to-Repository-to-PostgreSQL persistence, foreign-key integrity, unique constraints, soft-delete filtering (deletedAt), and transactional rollback behavior (prisma.$transaction).
  - Cross-module state transitions: Job Posting → Proposal Submission → Hiring Decision → Contract Generation & Dual Signing → Milestone Escrow Funding → Work Submission → Approval & Stripe Transfer.
  - External integrations: Stripe PaymentIntent/Checkout Session creation and raw-body webhook signature verification, Google OAuth token exchange, Cloudinary upload, SMTP email delivery, and Socket.IO WebSocket event broadcasting.
- **Acceptance Criteria:**
  - 100% of integrated API routes conform to the shared Zod request/response envelope and return accurate HTTP status codes.
  - Webhook handlers process duplicate Stripe events idempotently without creating duplicate ledger entries.

1.2.3 System Testing

- **In-charge:** Pham Kha Vy (Test Lead), Nguyen Tien Thuan, Tran Phat Tai, Vo Nhut Tin, and Nguyen Lu Nhat Huy.

- **Input / Time:** Software Requirement Specification (Report 3) and deployed staging containers; executed at the end of every sprint and during the Sprint 5 full regression cycle.
- **Focus:**
  - End-to-end validation of all functional use cases across Guest, Client, Freelancer, Expert, and Administrator roles through the Next.js web interface and backend APIs.
  - Negative authorization testing (role boundary enforcement, revoked session tokens, banned user restrictions), file upload validation, bilingual UI completeness (`en`/`vi`), and responsive layout behavior.
- **Acceptance Criteria:**
  - 100% of the 411 documented system test cases across all 14 functional modules in `Report5_Test Report.xlsx` pass.
  - Zero Critical or Major defects remain open, and `docker compose build` / `docker compose up -d` succeed with healthy containers.

1.3 Testing Constraints and Assumptions

Assumptions

- Third-party sandbox services (Stripe Test Mode API, Google OAuth 2.0 test credentials, Cloudinary media storage, and SMTP test relay) remain available and behave consistently with their documented contracts during test execution.
- The PostgreSQL test database is initialized from Prisma migrations and populated deterministically using the repository seed scripts (`seed`, `seed:permissions`, `seed:job-categories`, `seed:jobs-dataset`, and `seed:freelancers-dataset`) before executing system and regression test suites.
- External AI microservices for eKYC verification and CV skill extraction are reachable over HTTP (`FREVIA_AI_URL`) or simulated with deterministic test fixtures so that both automated scoring and administrative manual review fallbacks can be verified.
- Evaluators and UAT participants use modern desktop/mobile browsers (Google Chrome or Microsoft Edge) with camera permissions enabled when testing the eKYC selfie and liveness capture steps.

Constraints

- **Sandbox Payment Restriction:** All financial flows (platform fee payment, milestone funding, freelancer payouts, refunds, and dispute split settlements) must be executed strictly in Stripe Test Mode using test cards; real banking networks cannot be exercised.
- **Time-Dependent Business Rules:** Multi-day deadlines (such as proposal expiration, 72-hour dispute fee payment windows, and arbitration defence deadlines) cannot be waited on in real time and require controlled timestamp adjustments in the test database.
- **Document & Camera Hardware Dependency:** Automated eKYC OCR and liveness detection depend on physical camera quality and Vietnamese citizen ID card formatting; edge cases with damaged or low-light documents rely on curated image fixtures.
- **Single-Instance Deployment Topology:** Testing is conducted on a single container host with in-memory WebSocket room state, bounding load testing to single-node concurrency limits.

2. Test Strategy

2.1 Testing Types

2.1.1 Unit Testing

- **Objective:** Verify the internal logic, state validations, data transformations, and error handling of individual methods in NestJS services, repositories, guards, and utilities in isolation from external infrastructure.
- **Technique:**
  - **Equivalence Partitioning & Boundary Value Analysis:** Partitioning inputs into valid values, empty/whitespace strings, null/undefined parameters, pagination bounds (`page = 1`, `limit = 10`, maximum page size), skill proficiency limits (`1` to `10`), and file size limits (`10 MB`).
  - **White-Box Branch & Mock-Based Exception Testing:** Mocking `PrismaService`, `JwtService`, and external adapters via `jest.fn()` to exercise normal paths (`N`), abnormal exception branches (`A`), and boundary conditions (`B`).
- **Completion Criteria:**
  - 100% pass rate across all 507 unit test cases (`UF-01` to `UF-84`) documented in `Report5_Unit Test.xls`.
  - Minimum 80% code coverage on core business logic modules and zero lint or TypeScript compilation errors.

2.1.2 Integration Testing

- **Objective:** Validate that interacting controllers, services, repositories, Prisma database transactions, WebSocket gateways, and third-party adapters exchange data accurately and preserve relational integrity.
- **Technique:**
  - **Bottom-Up & Contract-First API Testing:** Testing repository classes against a real PostgreSQL instance first, followed by HTTP endpoint verification via Supertest and Postman against shared `@shared/types` Zod schemas.
  - **Transactional & Webhook Event Simulation:** Executing multi-table Prisma transactions and replaying signed Stripe webhook payloads to verify idempotency and ledger consistency.
- **Completion Criteria:**
  - 100% of API endpoints, database migrations, and webhook handlers execute without unhandled exceptions or schema mismatches.
  - All multi-actor state machines (Job, Proposal, Contract, Milestone, Dispute) persist accurate state across module boundaries.

2.1.3 System Testing

- **Objective:** Verify that the fully integrated Frevia web application satisfies all functional and non-functional requirements specified in the Software Requirement Specification (Report 3).

- **Technique:**
  - **Black-Box Use-Case & Scenario Testing:** Executing end-to-end test procedures across all 14 functional modules and 84 features from the user interface through to the database.
  - **Role-Based Negative & Exploratory Testing:** Exercising protected pages and API endpoints with different active roles (`GUEST`, `CLIENT`, `FREELANCER`, `ADMIN`) to verify RBAC enforcement, validation feedback, and empty/error UI states.
- **Completion Criteria:**
  - 100% of the 411 system test cases in `Report5_Test Report.xlsx` pass (`411/411 Passed`).
  - Zero open Critical or Major defects prior to Sprint Review and final release.

2.1.4 User Acceptance Testing (UAT)

- **Objective:** Confirm that the delivered platform meets business workflows, usability expectations, and domain requirements from the perspective of end users (Clients, Freelancers, Administrators) and the project supervisor.
- **Technique:**
  - **Incremental Sprint Reviews & End-to-End Walkthroughs:** Live demonstrations on the staging environment at the end of each sprint, followed by structured UAT scenarios in Sprint 5 with representative Clients and Freelancers completing full marketplace lifecycles.
- **Completion Criteria:**
  - At least 95% of core business workflows are accepted on the first walkthrough, with 100% of reported UAT feedback resolved before final submission.
  - Formal acceptance sign-off from the Product Owner (`Pham Kha Vy`) and Supervisor (`Cao Hong Nga`).

2.1.5 Non-Functional Testing

- **Objective:** Ensure that the system satisfies security, access control, query performance, bilingual localization, and container deployment reliability standards.
- **Technique:**
  - **Security & Authorization Testing:** Attempting privilege escalation (calling administrative routes with client/freelancer tokens, mutating another user's resources, replaying revoked session tokens, and sending unsigned webhook requests).
  - **Performance & Pagination Verification:** Verifying response times on indexed list endpoints (jobs, proposals, forum posts, users, skills, categories) and concurrent Socket.IO messaging sessions.
  - **Localization & Cross-Browser Testing:** Running automated i18n checks (`check-i18n` and `check-icu`) and validating UI rendering across Chrome, Edge, and Firefox in English and Vietnamese.
- **Completion Criteria:**
  - 100% of unauthorized requests are blocked with `401 Unauthorized` or `403 Forbidden`.
  - List endpoints respond within < 500 ms on staging, and 100% of English/Vietnamese translation keys are synchronized.

2.2 Test Levels

Table 1. Test levels table

Type of Tests	Test Level			
	Unit	**Integration**	**System**	**Acceptance**
Functional & Business Logic Testing	✓	✓	✓	✓
API Contract & Schema Validation	✓	✓	✓	
Database & Transaction Integrity		✓	✓	
Security & RBAC Authorization	✓	✓	✓	
UI/UX, Localization & Workflow			✓	✓

2.3 Supporting Tools

Table 2. Supporting tools table

Purpose	**Tool**	**Vendor/In-house**	**Version**
Unit & Integration Testing	Jest / `@nestjs/testing`	Meta / NestJS	30.0.0 / 11.0.1
API & HTTP E2E Testing	Supertest / Postman	Open Source / Postman	7.0.0 / 11.x
Static Analysis & Validation	ESLint / TypeScript / Zod	Open Source / Microsoft	9.18.0 / 5.9.2
Payment & Webhook Sandbox	Stripe CLI & Test Mode	Stripe Inc.	22.6.2
Test & Defect Management	MS Excel / GitHub Issues	Microsoft / GitHub	2024

3. Test Plan

3.1 Human Resources

Table 3. Human resources table

Worker/Doer	Role	Specific Responsibilities/Comments
Pham Kha Vy (CE182254) vypkce182254@fpt.edu.vn	Team Leader / Test Lead	Defines the test strategy and test plan; authors and executes unit tests (UF-01 to UF-36, 219 TCs) and system test cases (154 TCs) for Manage Forum, Conversation, Moderation, ID Verification, and Forum Trash; reviews all test deliverables and leads UAT.
Nguyen Tien Thuan (CE181024) thuanntce181024@fpt.edu.vn	Backend Developer / Functional Tester	Authors and executes unit tests (UF-37 to UF-84, 288 TCs) and system test cases (257 TCs) for User Authentication, Role Management, Manage Permission, Session Management, Forum Category, User Management, Banners, Skills, and Job Categories.
Tran Phat Tai (CE181632) taitpce181632@fpt.edu.vn	Full-Stack Developer / Integration Tester	Executes unit, integration, and system tests for Job Marketplace, Bookmarks, Saved Searches, Proposal workflows, and Hiring decisions; performs cross-browser UI validation.
Vo Nhut Tin (CE180619) tinvnce180619@fpt.edu.vn	Backend Developer / Security & AI Tester	Executes integration and security tests for Stripe Connect escrow payments, webhook signature & idempotency, dispute arbitration splits, AI eKYC verification, and AI CV skill extraction.
Nguyen Lu Nhat Huy (CE181196) huynlnce181196@fpt.edu.vn	Frontend Developer / System Tester	Executes integration and system tests for Socket.IO real-time messaging, notifications, freelancer portfolios, bilingual localization (en/vi), and responsive UI states.

3.2 Test Environment

Table 4. Test environment table

Purpose	Tool	Provider	Version
Backend Runtime & API	Node.js / NestJS	OpenJS / NestJS	22.15.x / 11.0.1
Web Client Framework	Next.js / React / Tailwind	Vercel / Meta	16.2.0 / 19.2.0
Database & ORM	PostgreSQL / Prisma	PostgreSQL / Prisma	16.x / 7.8.0
Container & Secrets	Docker Compose / Infisical	Docker / Infisical	27.x / CLI
Web Browsers	Google Chrome / MS Edge	Google / Microsoft	130+

3.3 Test Milestones

Table 5. Test milestones table

Milestone Task	Start Date	End Date	Comments
Test Planning & Setup	07/09/2026	19/09/2026	Define test strategy, configure Jest 30, Docker Compose, and Stripe sandbox (Sprint 1).
Sprint 1 Test Execution	21/09/2026	03/10/2026	Unit & system tests for Authentication, RBAC Roles, Profiles, Jobs, and Forum foundation.
Sprint 2 Test Execution	05/10/2026	17/10/2026	Unit & system tests for Proposals, Contracts, Milestones, Permissions, Sessions, and Categories.
Sprint 3 Test Execution	19/10/2026	24/10/2026	Unit & system tests for Messaging, Moderation, ID Verification, Trash, and User Management.
Sprint 4 Test Execution	26/10/2026	31/10/2026	Integration & system tests for Stripe Escrow, Webhooks, Disputes, Banners, and Skills.
Sprint 5 Regression & UAT	01/11/2026	04/11/2026	Job Category tests, full regression run (507 Unit TCs, 411 System TCs), and final UAT sign-off.

4. Test Cases

The unit matrix workbook **Report5_Unit Test.xls** now contains 178 function matrices: the original 121 matrices and 57 added matrices for Nhat Huy, with 582 newly executed cases. The system workbook **Report5_Test Report.xlsx** contains 817 case specifications across 39 module sheets, including 198 new cases marked Pending. The table below preserves the original 04 November baseline; current execution and extension figures follow in Section 5.

Table 6. Historical module distribution reported on 04 November 2026

No	Module Name	Functions	Unit TCs	System TCs	Tester in Charge
1	Manage Forum	17	103	69	Pham Kha Vy
2	Manage Conversation	8	48	35	Pham Kha Vy
3	Manage Moderation	3	18	14	Pham Kha Vy
4	Manage ID Verification	4	26	18	Pham Kha Vy
5	Manage Forum Trash	4	24	18	Pham Kha Vy
6	User Authentication	5	39	42	Nguyen Tien Thuan
7	Role Management	6	40	32	Nguyen Tien Thuan
8	Manage Permission	2	12	12	Nguyen Tien Thuan
9	Session Management	3	18	17	Nguyen Tien Thuan
10	Forum Category Management	6	36	29	Nguyen Tien Thuan
11	User Management	9	59	42	Nguyen Tien Thuan
12	Banners Management	5	23	24	Nguyen Tien Thuan
13	Skill Management	6	29	27	Nguyen Tien Thuan
14	Job Category Management	6	32	32	Nguyen Tien Thuan
Total		**84**	**507**	**411**	

5. Test Reports

5.1 Current Verification: Nhat Huy, 09 November 2026

The extension covers all 51 use cases assigned to Nguyen Lu Nhat Huy across Sprints 1–5, using the current Report 3 requirements, Report 4 designs, backend service code, controller DTOs and shared Zod schemas.

- **Unit test matrices:** 57 methods, `UF-122` to `UF-178`, containing 582 cases. Each matrix belongs to one method; dependencies are mocked. DTO validation cases are distinguished from service logic cases.

- **Executed unit tests:** all 582 added cases passed. The complete `test:unit` suite passed 1,358 cases across 34 suites. TypeScript checking completed successfully. This is a test execution result, not a measured statement-coverage percentage.
- **System/API specifications:** 198 added cases across 16 `NH` worksheets in the system test workbook. All remain **Pending**. Dates and testers are intentionally blank until execution. Unit mocks do not establish browser, authentication, RBAC, database, storage, or third-party integration correctness.
- **Existing material:** matrices `UF-01` to `UF-121` and the other members' system case rows are retained. Their original reported results are historical records; they were not inferred from the new cases.

Table 7. Nhat Huy added unit case distribution

Module	Cases	Normal	Abnormal	Boundary
Freelancer Profile	52	9	26	17
Freelancer Skills	27	4	19	4
Portfolio	65	14	37	14
Client Profile	20	6	11	3
Social Links	14	4	10	0
Favorite Freelancers	14	4	10	0
Following Freelancers	15	5	10	0
Notifications	8	5	2	1
Role Switching	12	4	8	0
Reviews	55	15	30	10
Review Responses	22	4	14	4
Account Profile	37	8	23	6
Profile Revision Review	44	15	19	10
Expert Profile	47	9	22	16
Identity Verification	33	8	23	2
CV Management	117	23	64	30

Implementation notes. Role names in access checks are `Client`, `Freelancer`, `Admin`, and `Expert`; revision type enums use uppercase values. CV upload and replacement share one service method and one matrix. Administrator revision history is provided through the status-filterable administrator list; the account endpoint returns only the latest revision. Uploaded-file size errors raised by Multer are HTTP 413, while direct service validation exceptions can be HTTP 400. These layers must not be conflated.

Evidence and reproduction. The unit matrix and test source share a catalogue in `tools/nhuy-test-catalog.json`. The detailed mapping, commands, execution scope and pending verification are documented in `README-NHUY.md`. No production

code was changed for this extension.

5.2 Historical Results Reported on 04 November 2026

The remaining tables and analysis below preserve the original report's baseline figures. They are not the execution totals of the 09 November extension and do not establish completion of the newly added system test specifications.

5.3 Test Execution Summary

Table 8. Test execution summary table

Test Type	Total Test Cases	Passed	Failed	Pass Rate
Unit Testing	507	507	0	100.00%
Integration Testing	84	84	0	100.00%
System Testing	411	411	0	100.00%
Acceptance Testing	42	42	0	100.00%
Total	**1,044**	**1,044**	**0**	**100.00%**

5.4 Defect Summary

Table 9. Defect summary table

Severity	Number of Defects	Description
Critical	0 open (2 resolved)	Early sprint defects involving duplicate Stripe webhook event delivery and revoked refresh token reuse were resolved by adding idempotency checks and strict session state validation (100% resolved).
Major	0 open (6 resolved)	Edge cases in soft-deleted category/skill slug uniqueness, role-switching permission lookup, and milestone state transition guards were fixed and verified during Sprints 2–4 (100% resolved).
Minor	0 open (9 resolved)	UI validation message formatting, missing Vietnamese (`vi`) translation entries, and pagination filter reset issues were resolved prior to the Sprint 5 regression run (100% resolved).

5.5 Test Analysis

- **Unit Testing Coverage and Distribution:** All 507 unit test cases across 84 service functions (`UF-01` to `UF-84`) passed (`100.00%` pass rate). The suite thoroughly exercises 240 normal scenarios (`47.34%`), 196 abnormal/exception paths (`38.66%`), and 71 boundary conditions (`14.00%`), exceeding the 80% code coverage quality gate defined in the Project Management Plan.
- **System and Integration Stability:** All 411 system test cases across the 14 functional modules in `Report5_Test Report.xlsx` and 84 integration workflow checks passed with `100.00%` test coverage and `100.00%` successful execution coverage on the containerized staging environment.
- **Defect Prevention via Shared Contracts:** Sharing Zod schemas (`@shared/types`) between the Next.js 16 frontend and NestJS 11 backend, combined with centralized RBAC enforcement in `AccessTokenGuard`, prevented client-server payload drift and eliminated unauthorized access regressions across all five sprints.
- **Defect Resolution Rate:** All 17 defects discovered during iterative sprint testing (2 Critical, 6 Major, 9 Minor) were fixed, covered by regression tests, and closed within the sprint in which they were detected, leaving zero open defects at release.

5.6 Test Conclusion

On 09 November 2026, all 1,358 executable backend unit tests passed, including 582 added cases for Nhat Huy's 51 use cases. The new matrix specifications and 198 system/API scenarios are complete as test documentation. System/API execution remains pending; no release-readiness or full integration claim is made from the unit results. The earlier figures above are retained as the original project's historical report.